

SAAD FAROOQ

Electrical Engineering | Computer Engineering | Microprocessors & Robotics
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EDUCATION

New Jersey Institute of Technology (NJIT)

2025 - Present

B.S. Electrical Engineering; Minor in Computer Engineering; Concentration Computer Systems | Albert Dorman Honors College

- Honors scholarship covering 100% of tuition and housing.
- Current independent study includes Physics II and Calculus II.

ACADEMIC PREPARATION

Mathematics: A Level Chemistry, A Level Mathematics, A Level Physics; SAT Math: 770/800.

Research interests: semiconductor devices, energy systems, power electronics, embedded systems, sensing and instrumentation, energy-efficient engineering, and renewable energy technologies.

TECHNICAL SKILLS

Programming: Python, MATLAB | **Hardware:** Arduino | **Engineering:** CAD, UAV Design, Data Analysis | **Remote Sensing:** MODIS, TRMM, GPM

PROJECTS

FloodShield: Flood Resilience Sensor System

Arduino | TRMM & GPM | Environmental Sensing

- Co-developed a <\$10 waterproof Arduino flood sensor; tested housing and calibration in flood-affected communities.
- Integrated TRMM/GPM rainfall data with ground sensor thresholds to assess flood-risk relationships.

Satellite-Based Flood Prediction

MODIS | Sentinel | Remote Sensing

- Analyzed MODIS/Sentinel spectral data and rainfall records to identify early flood indicators in South Asian river basins.
- Developed a rainfall-to-flood correlation approach with ~38% improvement in validation accuracy.

Morphing Wing UAV Research

CAD | NACA Airfoils | Aerodynamics

- Compared servo, linear, and ball-screw actuators for morphing UAV wings across force, power, and cost metrics.
- Modeled actuator-airfoil combinations using NACA profiles and produced CAD designs for a 5,000-word research study.

INDEPENDENT STUDY & RESEARCH PREPARATION

- Exploring semiconductor device physics, electronic materials, and energy conversion to build a foundation for undergraduate research in semiconductor and energy systems research.
- Strengthening electrical engineering fundamentals through independent study of circuit analysis, electronics, and mathematical methods relevant to semiconductor devices and power systems.

ACTIVITIES

NJIT Baja Racing - Electrical Team

- Contributing to electrical systems for an off-road race vehicle and developing hands-on exposure to hardware and electromechanical systems.

NJIT Autonomous Racing Club | IEEE Hardware Project Sprint

- Participating in campus robotics, autonomy, and hardware-focused engineering activities.

AWARDS & LEADERSHIP

- **Pakistan Young Physicists Tournament** | Team Captain, 2024
- **International Astronomy & Physics Competition** | Pakistan Team Member, Bronze Medalist
- **President, Environment Society**- 2023-2025.
- **International Chemistry Tournament** | Pakistan Team Member